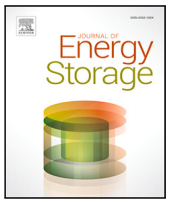
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Research papers



Understanding lithium-ion battery degradation in vehicle applications: Insights from realistic and accelerated aging tests using Volkswagen ID.3 pouch cells

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ABSTRACT

In this paper, we provide an in-depth analysis of long-term aging tests of over three years of continuous cycling using lithium-ion battery cells of the *2020 Volkswagen ID.3 Pro Performance*. The study examines the degradation of state-of-the-art NMC/graphite pouch cells and compares it with the degradation behavior in the field application under real-world conditions. The cell level tests include two realistic edge-case scenarios, namely a long-distance highway driving pattern as well as a short trip urban-interurban commuter pattern, alongside a synthetic, accelerated 1C/1C test case. Identifying the related dominating degradation modes reveals four key findings: first, the battery cells demonstrate exceptional cycle life performance, achieving nearly 400 000 km and 1100 fast-charge cycles between 20 % and 80 % state of charge in the highly demanding edge-case scenario. Second, accelerated synthetic load conditions significantly reduce testing time, with 1C/1C constant current cycles covering the OEM's cycle aging warranty of 160 000 km within 14 weeks, compared to 24 weeks for the highway and 1.5 years for the commuter driving scenarios. Third, a notable mismatch in the degradation behavior of the three examined test cases is observed, with hardly any degradation of the negative electrode and only a slight loss of lithium inventory in the commuter scenario. In contrast, the elevated stresses acting in the highway scenario and 1C/1C cycling cause a considerably higher loss of lithium inventory in combination with severe degradation of the negative electrode, ultimately resulting in a knee point. Fourth, inhomogeneous lithium distribution in the negative electrode is found due to omitted rest periods, depending on the stress level. This inhomogeneous lithium distribution further limits the transferability of synthetic accelerated laboratory test results to real-world vehicle applications, as these tests do not adequately replicate the degradation behavior observed in field applications. Nonetheless, a high level of agreement between the cell test applying the commuter driving pattern and the field application could be proven.

1. Introduction

Between 1832 and 1839, Robert Anderson presented the first battery electric vehicle (BEV) [1,2]. Since then, BEVs have disappeared from the scene, experienced a revival, and the technology inside underwent several evolution cycles, putting forth lithium-ion batteries (LIBs) in the early 1990s [3], which now have become the key technology for traction batteries. However, one crucial challenge persists: battery lifetime performance. As several studies reveal [4,5], customers are still concerned about the longevity of the traction battery. Similarly, OEMs are cautious in specifying battery warranties, with coverage rarely exceeding 250 000 km or ten years [6,7].

In fact, the impact of aging in real-world vehicle operation has been shown in a few field and vehicle studies [8–11]. Inevitably occurring aging mechanisms deteriorate the battery performance, and therefore, lifetime is limited [12]. Those aging mechanisms are triggered by the operating conditions, also called stress factors, to which the battery

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is exposed throughout its service life. As a consequence, the stress factors cause aging effects, becoming apparent in capacity loss and resistance increase [13–18]. Both occur during active usage as well as during inactive sequences. The former includes charging and discharging processes, known as cycle aging, whereas the latter refers to the idling condition of the battery while the vehicle is parked, thus referred to as calendar aging. They have been extensively investigated under synthetic laboratory conditions [15,16,19–21]. For example, high charge rates are known to promote lithium plating at the negative electrode [22], particularly in combination with low temperatures and high SOCs [12,23], whereas increasing depths of discharge (DODs) cause growing mechanical stresses due to intensified volume expansion, ultimately resulting in particle cracking of the active electrode materials [24], especially for graphite blend materials with high silicon shares [25]. Moreover, high temperatures and SOCs exacerbate transition metal dissolution at the positive electrode [26] as well as electrolyte decomposition [16]. While these aspects are only a few examples, they illustrate the diverse impacts and complex interdependencies of different stress factors on battery aging, in particular when considering the wide range of operating conditions of vehicle applications.

In order to select a suitable battery cell and to ensure the lifetime requirements, the cells are subjected to extensive aging studies before being integrated into a vehicle [18,27,28]. However, testing time is scarce due to fast and time-constrained development cycles in the automotive sector [29] and in the fast-paced field of battery technology [3]. Hence, battery aging tests are primarily performed in an accelerated manner [27,30–37]. As a result, those accelerated aging studies commonly involve abstracted and synthetic load conditions such as constant current (CC) load cycles [21,24,27,31–34,38,39].

Considering the *Volkswagen ID.3* with a battery warranty of 160 000 km or eight years (whichever is reached first) and an energy consumption of 0.75 Wh/km converted to the cell level in the WLTC based on the 9s(12s2p) module-cell configuration [40], an overall energy throughput of 120 kWh needs to be covered. Given an energy content on cell level of 280.6 Wh [40], this equals approximately 428 EFCs, whereas one EFC comprises one complete charge and discharge cycle of the cell normalized to the nominal capacity. In contrast to eight years of service life, this takes around 36 days of testing time with 1C/1C full cycles omitting constant voltage (CV) phases. Furthermore, performing intermediate check-ups (CUs) of, e.g., two days of testing time every 100 EFC, adds another ten days to the overall test duration.

1.1. State of the art

As the sample calculation shown above highlights, accelerated aging tests aim to reduce the necessary testing time of several years of service life to a few months or even weeks. This is commonly achieved by applying elevated stress levels compared to an unaccelerated reference case. Elevated stresses aggravate degradation effects, thus increasing the degradation rate. Most accelerated aging tests come along with an aging model to translate and extrapolate the measured aging behavior in the laboratory into a corresponding lifetime projection for the considered application. For further details, we refer to Paarmann et al. [13], who reviewed the literature focusing on the influence of various stress factors on aging behavior.

In many cases, extensive aging studies with large test matrices of different stress factor combinations are performed, varying one factor at a time. This strategy intends to understand the sensitivities of degradation effects regarding individual stress factors [24,31,32,41–43]. As a promising degradation acceleration factor, elevated operating temperature has been identified [27,34,41,44]. For extrapolation and lifetime projection of the measured aging data, empirical [27,32,34,44–46] or semi-empirical aging models [35–37,47–52] are often employed. As an alternative, machine learning [53,54] or physics-based aging models [55] are occasionally used.

However, it is claimed that no sound conclusions from these accelerated aging studies under synthetic operating conditions can be drawn about the degradation effects that occur in unaccelerated real-world applications [56–59]. Lewerenz et al. [60] and Morales Torricos et al. [61] conducted accelerated aging tests under various elevated C-rate and DOD combinations and found considerable shares of reversible aging effects and, thus, notable capacity recoveries. Therefore, they recommend performing rest phases during or at the end of accelerated aging tests, which is omitted in the magnitude of aging studies. Moreover, path dependency has been identified to significantly impact the aging behavior of LIBs [57,62–65], especially when comparing the monotonous test characteristics in the laboratory with the highly versatile and dynamic operating conditions in the vehicle application. This is further emphasized by the findings of Anséan et al. [58] demonstrating an impact of quickly alternating de-/intercalation processes on aging behavior, which are characteristic for real-world vehicle operation in contrast to constant current cycling.

Hence, more realistic load profiles are gaining increasing interest, e.g., for aging investigations or model validation. However, most of these studies utilize generic, accelerated load cycles [66,67] or standard drive cycles such as the new european driving cycle (NEDC) [52], the WLTC [49,68,69], the federal urban driving schedule (FUDS) [57], the urban dynamometer driving schedule (UDDS) [70], or other synthesized drive cycles [57,71–73]. Baure et al. [57] show that such synthetic cycle conditions are incapable of reproducing real aging behavior by comparing aging data measured under different standard drive cycles in the laboratory with aging data measured in public road traffic. The authors highlight the importance of proper consideration of operating conditions, especially due to path dependence in battery aging trajectories. The impact of real-world load profiles has scarcely been investigated so far, providing only a few studies [57,74].

Summarizing the state of the art discloses a mismatch between established methods of battery aging tests and the real-world application. It is therefore questionable how meaningful synthetic, accelerated test conditions are when assessing the highly dynamic and diverse vehicle operation. Additionally, it is uncertain whether and how the lifetime prediction from accelerated aging tests can be transferred to the real-world application.

1.2. Contributions

From the presented state of the art, several questions arise. Due to the lack of accessibility to extensive field and experimental aging data under realistic conditions, it has not yet been conclusively clarified how exactly automotive-grade battery cells age under different real-world operating scenarios. Furthermore, it is not yet known which degradation modes and aging mechanisms dominate the degradation behavior and how the user is impacted in daily operation. Moreover, it remains unclear how the measured aging behavior in the laboratory matches the degradation in the vehicle operation.

To fill the above-mentioned research gaps, we assess the battery aging of large-scale state-of-the-art automotive-grade pouch cells from the *Volkswagen ID.3 Pro Performance* (manufacturing year 2020) in a cycle aging study under realistic and accelerated conditions. The key characteristic of our approach is the comparability between laboratory and vehicle levels, closing the aforementioned gap between testing and field application. This is achieved by transferring the operating conditions from the field application to the laboratory, where an identical cell is examined as the one installed in the test vehicle. In this work, we consider different perspectives, as depicted in Fig. 1:

(1) Laboratory perspective: transferability between accelerated and realistic aging tests.
The degradation behavior is investigated on cell level. Based on that, the transferability between accelerated and unaccelerated conditions is evaluated through capacity and resistance analysis, DVA and incremental capacity analysis (ICA), as well as the examination of degradation modes.

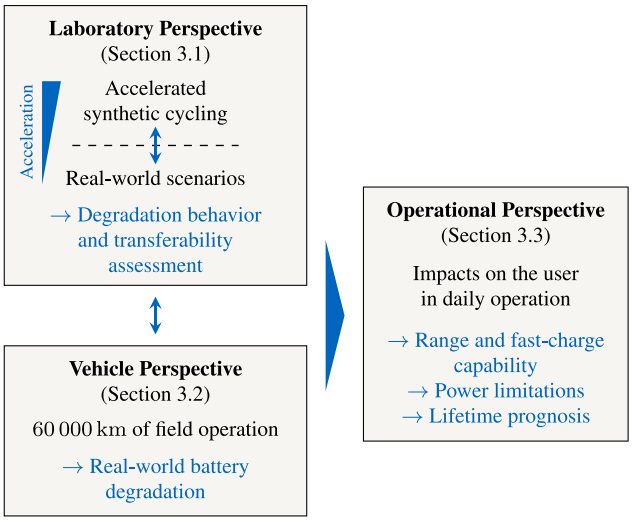


Fig. 1. Article layout including three investigation perspectives and the associated research topics.

(2) Vehicle perspective: degradation analysis under real-world conditions.
We investigate the battery pack aging in the vehicle under the influence of real-world operating conditions as well as system impacts. Using DVA, the degradation behavior is compared with cell level aging.

(3) Operational perspective: user impacts due to battery aging.
We shed light on the impacts of battery aging on the vehicle application and how the user is affected in daily operation. For this purpose, we analyze the influences on range and fast-charging capabilities and evaluate the resulting power limitations. Finally, we derive a lifetime prognosis from the combined findings of both the accelerated laboratory tests and the real-world aging behavior in the vehicle.

2. Experimental

The following section describes the investigated battery cell and vehicle under study. Thereafter, the measurement equipment and the test design are explained, followed by the applied data analysis procedures.

2.1. Investigated specimens

Battery cell under study – The aging studies are conducted on a state-of-the-art 78 Ah NMC/graphite pouch cells (manufactured by LG Energy Solution) out of the *Volkswagen ID.3* as characterized by Günter and Wassiliadis [75]. The tested cells were disassembled from a series production vehicle at a mileage of a few thousand kilometers. The cells' history up to that point is not known in detail but is assumed to be negligible to the underlying aging study as the mean initial cell capacity amounts to 79.4 Ah with a standard deviation of 0.06 Ah. Before the start of the experiments, the cells were stored at 3.6 V at 20 °C for seven months. Table 1 summarizes the cell specifications.

Vehicle under study – The vehicle under study is an unmodified *VW ID.3 Pro Performance* containing the midsize 58 kW h (net) battery manufactured in 2020. The battery pack comprises nine modules, each containing 24 battery cells in an overall 108s2p layout [40]. The test vehicle had a mileage of around 400 km at the beginning of the study and approximately 60 000 km at the end of the considered test period. In between, it underwent a stochastical usage pattern as a company car, including short and long trips in urban, interurban, and highway driving conditions, as well as occasional fast-charging.

Table 1
Specifications of the investigated cell specimen. The listed data is based on our own and previous measurements [40,75]. An official datasheet cannot be provided, as the cell is an automotive-grade cell extracted from a real vehicle and detailed cell specifications are not disclosed by most vehicle manufacturers. However, based on the cell properties, we assume it is the model *E78* [76].

Parameter	Value
Manufacturer	LG energy solution
Cell type	E78 (presumably)
Cell format	Pouch
Negative electrode material	Graphite
Positive electrode material	NMC
Capacity (measured @ C/3 and 25 °C)	78 Ah
DC resistance (measured @ 1C, 10 s and 20 °C)	1.8 mΩ
Voltage range	2.5 V–4.2 V
Energy	300.6 W h
Cell dimensions	545 × 98 × 8.65 mm
Cell weight	1.1 kg

2.2. Measurement equipment

The cell testing is conducted using a *MRS-6V* battery test system (BaSyTec GmbH, Germany). A VC^3 *4100* temperature chamber (Vötsch GmbH, Germany) is used for thermal control. However, no active cooling is performed, as implemented in the vehicle via the bottom cooling plate. Instead, the cells are kept at a constant operating temperature. Each cell's surface temperature was observed with two PT100 thermal sensors close to the tabs for safety reasons. In order to preserve the mechanical compression on the cells, the tested specimens were kept inside the original module housing. For electrical contact, the housing's lids, as well as the cell and battery management system (BMS) connectors, were removed so that the cell tabs were individually accessible. The connection is realized by two copper bars clamped onto the cell tabs with two bolts each, as described in our previous study [40].

Data on the vehicle level was acquired by an in-house developed data logger connected to the OBD-II interface [40,77]. The relevant parameters were extracted through unified diagnostic services queries from the controller area network communication.

2.3. Test design and measurement procedures

The test design consists of alternating CUs and continuous cycling sequences, such as one of two real-world driving cycles or a synthetic profile. All measurements were performed at a constant ambient temperature of 20 °C, either in a climate chamber (cell level) or in an air-conditioned workshop (vehicle level). For each test case, one cell is investigated.

Cell check-up procedure – A standardized CU procedure is performed initially and regularly throughout the aging study. This CU consists of three main sequences that are separated by 15 min breaks. Initially, the capacity is determined with a C/3 constant current constant voltage (CCCV) discharge between 4.2 V to 2.5 V with a cut-off current of C/50. Subsequently, the pOCV is measured at C/15 in charge and discharge direction in the same voltage range. A hybrid pulse power characterization (HPPC) with CC pulses at 10%, 50%, and 90% SOC completes the procedure. From that, the DC pulse resistance is derived and evaluated for the 20 s, 1 C pulse at 50% SOC in charge direction.

Cell cycle aging test procedures – The real-world scenarios represent two opposing edge cases in terms of the stress level. The corresponding voltage and power profiles are provided in Appendix B. On the one hand, a short trip commuter pattern with urban and interurban driving conditions and an 11 kW alternating current (AC) wallbox charging marks a low demanding mild load case. The DOD is set to 8% (80% to 72% SOC) which corresponds to a trip length of 28 km. The maximum currents on cell level in the driving operation

lie between −133.1 A (−1.7 C) during discharging and 88.4 A (1.1 C) during regenerative braking. In the charging sequence 12.1 A (0.2 C) are reached. The corresponding test procedure is provided in Table A.1 and visualized in Fig. B.1(a).

On the other hand, a highly dynamic long-distance highway driving pattern with a 100 kW DC fast-charging replicates a highly demanding load case. Cycling is performed between 100% and 20% SOC followed by the DC fast-charging event up to 80% SOC and a second discharge to 20% SOC accumulating to a trip length of 375 km. Every second charging sequence is performed as 11 kW AC wallbox charging with a subsequent CV phase representing slow overnight charging while simultaneously serving as recalibration measure to avoid an SOC drift during cycling. The maximum currents occurring are −190.0 A (−2.4 C) during driving and 135.5 A (1.7 C) during regenerative braking, whereas 121.2 A (1.6 C) are reached during fast-charging. The corresponding test procedure is provided in Table A.2 and visualized in Fig. B.1(b). The applied DC fast-charging profile, depicted in Fig. B.2, is implemented as a current step profile, which switches to the next current step plateau as soon as predefined voltage thresholds are surpassed. The voltage thresholds are derived from a fast-charging measurement with the test vehicle to approximate the real charging profile [40]. This method assures a steady derating of the fast-charging current depending on the aging state to safely operate the cell and avoid overcharging.

Both scenarios are implemented as power profiles derived from the measured drive cycle on the dynamometer. Using the power demand instead of the measured current better replicates the real operating conditions since the load is then independent of the cell state, which varies throughout the aging process. For further details on the corresponding velocities, power profiles, currents, voltages, SOCs as well as a load spectrum analysis of the particular load regimes, we refer to our previous publication [40]. In addition, a synthetic load cycle was investigated, consisting of 1 C (78 A) CCCV charge and CC discharge cycles with 10 min rest in between for thermal relaxation. The voltage boundaries were chosen following the voltage window applied in the vehicle, ranging from 3.25 V to 4.19 V corresponding to the SOC window available by the vehicle BMS [40]. The corresponding test procedure is provided in Table A.3 and visualized in Fig. B.1(c).

For the commuter scenario, the CU is repeated after every 200th cycle, for the highway scenario after every 50th cycle, and for the 1C/1C cycle after every 100th cycle in order to avoid excessive deviation of the charge throughput during cycling.

Vehicle pOCV-measurement – The vehicle level pOCV is acquired by a charging test with a commercially available *Juice Booster* (Juice Technology AG, Switzerland) connected to a single-phase socket, as described in [40,78]. During this charging test, the vehicles are fully charged starting at 0% BMS SOC using a constant power (CP) charging profile, without a CV phase taking place as soon as the upper voltage limit is reached. The *Juice Booster* is set to the lowest possible current configuration of 8 A to minimize the influence of overpotentials. Additionally, the charging current is further reduced in the vehicle settings so that an approximately constant power charging of 1.2 kW is applied, leading to an approximate mean C-rate of $C/48$. Further details are given by Bilfinger et al. [78].

2.4. Data analysis procedures

Subsequently, the data analysis procedures used to process the measured data are presented. These procedures include the computation of the DVA and the ICA as well as the extraction of degradation modes.

***Differential voltage and incremental capacity* analysis** – DVA and ICA are used to assess the cells' degradation behavior on the electrode level. As the DV curve gives insight into characteristic intercalation stages of the lithium ions in the electrode structures, it reveals the thermodynamic processes and internal condition of the cell. Thus, distinct electrode-related features can be distinguished by specific peaks [79,80]. The DV is extracted from the pOCV measurement in charge direction as the derivative of the voltage in relation to the charge, as $f(Q) = \mathrm{d}U/\mathrm{d}Q$. Vice versa, the ICA is defined as the inverse, exhibiting the phase transitions in the electrodes. The measured input values, as well as the calculated output values, were smoothed using the Savitzky–Golay filter method. For further information regarding the numerical methods and implementation of the DVA, we refer to Bilfinger et al. [78]. The software tool applied in this study was developed in-house and is available as an open source [81]. The same software was used to process cell and vehicle data to avoid numerical and data post-processing impacts.

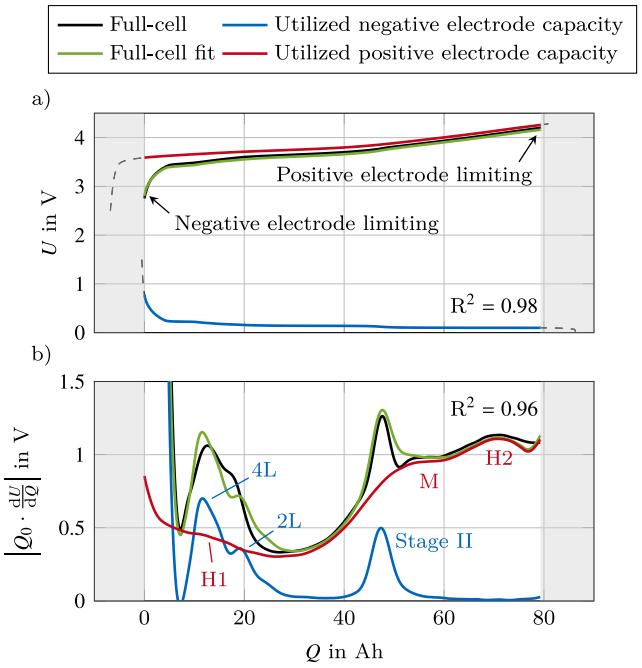


Fig. 2. Half-cell alignment of the investigated battery cell: (a) negative (blue) and positive electrode (red) pOCV curves as well as the resulting full-cell fit (green) in comparison to the measured full-cell pOCV (black). Note that the half-cell measurements have been conducted with a C-rate of C/50, whereas the full-cell measurement was performed with C/15, explaining the slight offset between the fit and measured curve. (b) alignment of the corresponding half-cell DV curves in relation to the full-cell DV. The half-cell data has been measured by Günter and Wassiliadis [75].

Fig. 2 visualizes the half-cell alignment and the calculated full-cell fit in comparison to the measured full-cell curve. The negative electrode constrains the full-cell potential at the end of discharge, whereas the positive electrode is limiting at the end of charge. Note that the half-cell pOCV measurements, taken from Günter and Wassiliadis [75], were performed at C/50 charge rate, while the full-cell pOCV measurements are carried out at C/15. This causes different cell polarization, resulting in an offset between the measured and the fitted pOCV curves. Since this effect is independent of the charge, it does not occur in the DV curves. For this reason, the DV is used as target function for the half-cell curve fitting. Moreover, the different charge rates explain the slight differences in the peak shapes, with higher currents blurring the peaks more and making them less pronounced. This becomes particularly apparent for the liquid-like graphite stages in the low SOC range.

Degradation mode analysis – By assessing the changes in the anodic and cathodic DV features throughout the cycle life testing, degradation modes can be quantified. These degradation modes include loss of active material (LAM) and LLI. LAM can further be distinguished into $\mathrm{LAM_{NE}}$ at the negative and $\mathrm{LAM_{PE}}$ at the positive electrode. The degradation modes are determined and interpreted as presented in literature [15,20,56,82,83]. For this purpose, the pristine half-cell potential curves are mathematically fitted to the full-cell curves measured in the CUs, yielding the scaling and shifting factors of both electrodes. Based

on these factors with respect to the initial condition, the degradation modes can be calculated, as explained by Wildfeuer et al. [56]. In the DVA, electrode degradation becomes apparent by a compression of the DV curve of the respective electrode. In contrast, the accessibility of lithium inventory, and from that, LLI can be evaluated by analyzing the electrode slippage, meaning the shift in electrode potential between positive and negative electrodes relative to each other [20,21]. The degradation modes are determined and interpreted as presented by Wildfeuer et al. [56].

3. Results and discussion

In the following, the results of the cell level measurements, as well as the vehicle level tests, are presented and discussed. Subsequently, the results are interpreted from an operational point of view.

3.1. Laboratory perspective

According to Attia et al. [19], different aging pathways can be distinguished depending on the applied loads and the resulting electrochemical degradation phenomena. In the following, we assess the occurring degradation modes and aging mechanisms under the applied operating conditions and identify the dominating phenomena on the cell level in the laboratory environment. For this purpose, we first consider the apparent aging effects by means of capacity fade and resistance increase. From that, we secondly investigate the underlying degradation behavior via DVA and degradation mode analysis.

3.1.1. Aging effects

With the progressive cycling of the examined cells, aging effects become apparent, as depicted in Fig. 3. The well-known sublinear capacity loss [35,37,48,84] is observed at the beginning of the test, characterized by an initial drop followed by a decrease in the degradation rate across all three test cases. Overall, the trend in resistance increase correlates well with the capacity loss of all three cells. The initial decline in capacity can be partially attributed to compensatory effects of the negative electrode's overhang, which amounts to approximately 2.5% according to the data provided by Günder and Wassiliadis [75]. Given that in all test cases, the mean SOC during cycling exceeds the storage SOC before testing, the overhang is charged during the first cycles [60,85].

In addition to the anode overhang effects, initial aging effects also result from post-formation processes such as solid electrolyte interphase (SEI) growth [21,26,36–39,60,84,86,87]. This aligns well with the underlying stress factors, where increasing DODs and higher C-rates contribute to an enhanced initial capacity loss and resistance increase. Literature suggests that higher loads, particularly DOD, correlate with increased damage and reformation of the SEI, which is directly related to volume expansion of the negative electrode during charging and discharging [39] and therefore, responsible for surface cracks in the active material [88]. The DOD shows a coincident impact on the initial degradation effects in our tests, with the most severe extent for the 1C/1C cycling, which has the highest DOD, followed by the highway and the commuter scenario. Furthermore, the initial internal resistance increase during the first 60 EFC confirms the occurrence of electrochemical degradation processes in the form of cover layer formation and growth at both electrodes [12,39,41,89–91]. After approximately 50 EFC in the commuter scenario and about 200 EFC in the highway scenario, as well as for the 1C/1C cycling, a nearly linear behavior ensues. In this linear aging phase, the load-dependent aging rate becomes apparent with the commuter and highway scenarios aging approximately at a comparable pace, while the 1C/1C cycling induces significantly faster degradation.

Within the linear aging phase, intermittent capacity increases are observed for all three cells. These capacity recoveries coincide with unintended pauses during cycling due to equipment failures, particularly pronounced around 250 and 550 EFC for the commuter scenario and between 500 and 750 EFC for the highway scenario. These capacity increases indicate cell relaxation occurring during the breaks, most likely resulting from reversible capacity losses during cycling, as reported by, e.g., Morales Torricos et al. [61]. Additionally, for the highway scenario, we suspect an influence of the negative electrode's overhang during the pause around 600 EFC, where the cell was idling at a comparably low voltage of 3.48 V, and consequently discharging the overhang.

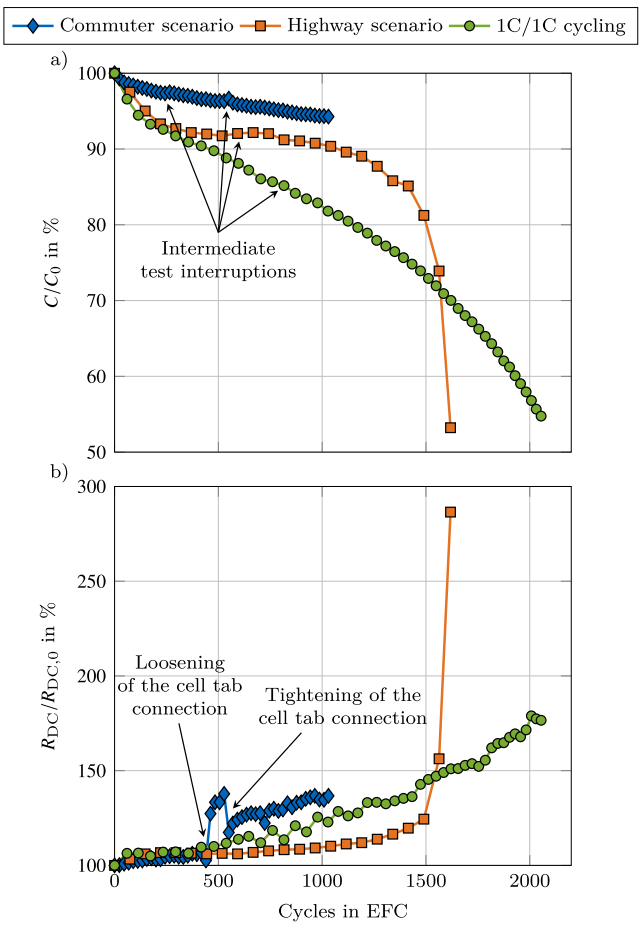


Fig. 3. Cycle life performance of the examined battery cells: (a) capacity loss, (b) resistance increase, both presented versus cycles in EFC.

The commuter scenario reveals an intermediate resistance step between around 450 and 550 EFC. This incident can be traced back to a disrupted electric contact due to the copper bars connected to the cell tabs becoming loose, thus increasing the contact resistance. After reattachment, a significant resistance reduction is observable. However, comparing the magnitudes before and after this incident shows that the resistance level is still higher, presumably because of an altered electric contact compared to the start of the experiment. This effect has been reported in literature [92].

Around 1400 EFC, the highway scenario exhibits a knee point, followed by a rapid capacity decline to approximately 52% at just over 1600 EFC. At this point, the experiment was terminated for safety reasons. We attribute this knee point to increased lithium plating due to the continuously growing SEI, leading to pore clogging [36,93]. This effect increases the charge transfer resistance [39] and simultaneously slows down reaction kinetics [26], thus raising the lithium plating propensity. The corresponding resistance growth to almost 300% supports this hypothesis [12], as it occurs in good agreement with the

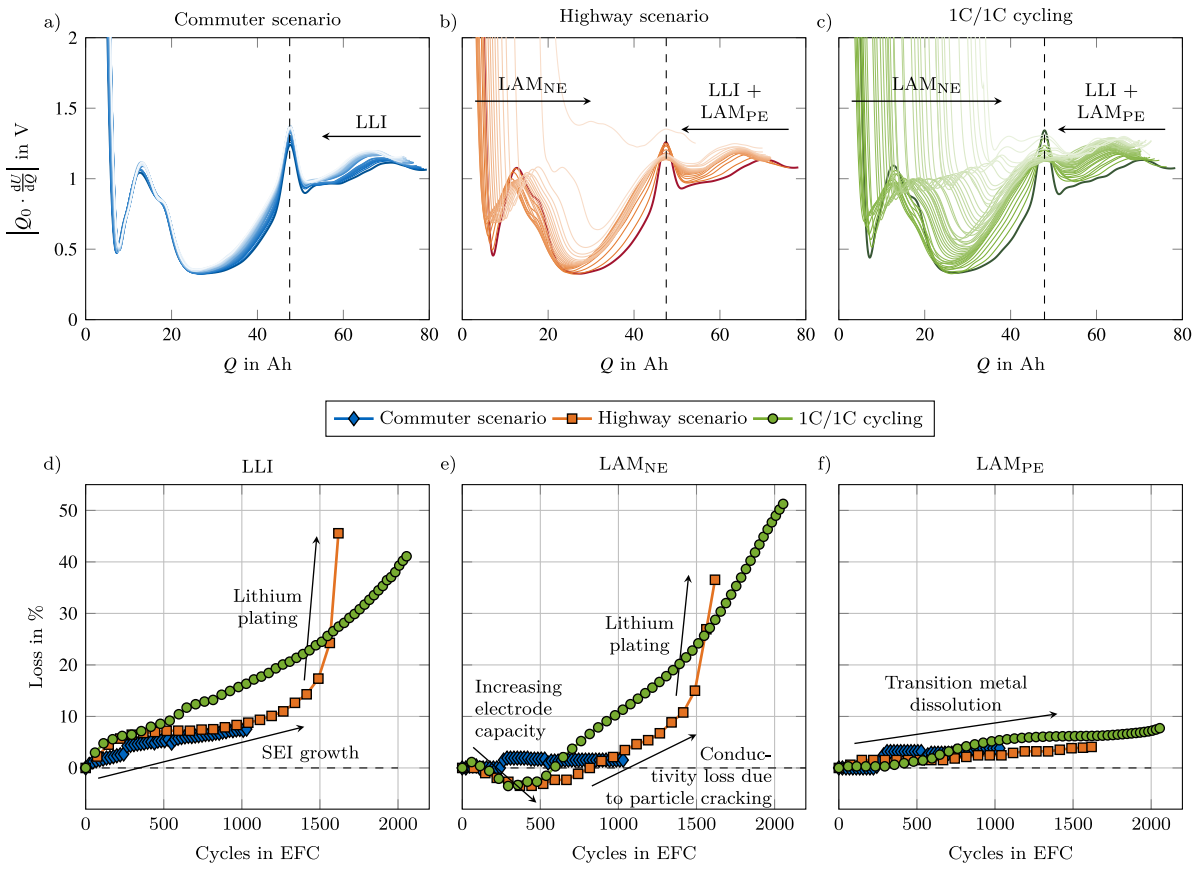


Fig. 4. Differential voltage and degradation mode analysis of the examined test cases: (a) commuter scenario, (b) highway scenario, (c) 1C/1C cycling, from dark (pristine condition) to bright (aged condition). For better evaluation of the occurring degradation modes, the curves are aligned by the graphite stage II feature indicated by the dashed line. (d) LLI, (e) LAM_{NE}, (f) LAM_{PE}, extracted from the DV curves by half-cell fitting. The losses are normalized individually for each degradation mode and do not add up to the overall capacity losses of the investigated cells.

knee point due to metallic lithium depositions on the negative electrode passivating areas of its active surface [50,51]. This elevated lithium plating propensity in the highway scenario is further increased by the frequently repeated fast-charging events, with the cell undergoing approximately 1100 fast-charge cycles from 20 to 80% SOC, with peak C-rates reaching around 1.6 C.

3.1.2. Degradation behavior

The DVA gives deeper insights into the diverse degradation effects under the varying test scenarios. Figs. 4(a)–(c) visualize the DV curve spectra for the three test scenarios. For better differentiation of the occurring effects, the DV curves are aligned by the graphite stage II, highlighted by the dashed line, as presented in [94]. Given the fact that the negative electrode is limiting at the end of discharge, alterations of the DV curves to the left of the graphite stage II can be directly attributed to LAM_{NE}. In contrast, due to the limiting impact of the positive electrode at the end of charge, changes to the right of the graphite stage II represent the superposition of LLI and LAM_{PE} [94]. ICA is used to validate the results and to better visualize these two degradation modes based on the methodology introduced by Dubarry and Anséan [95]. The respective visualizations are given in Fig. C.3 in Appendix C. Moreover, to assess and compare the degradation paths throughout the experiment, the degradation modes are quantified from the DV curves by half-cell fitting and presented in Fig. 4(d)–(f), revealing the underlying degradation processes.

The cell subjected to the commuter scenario exhibits only minor degradation overall. However, a discontinuous step in all three degradation modes around 250 EFC is conspicuous. At this point, the measurement was switched to another test channel due to a malfunction of the initial one. Considering this effect and taking into account that LAM and LLI remain unaltered apart from it, they presumably do not present actual physical degradation effects and can thus be neglected. Consequently, the dominating effect in the commuter scenario is LLI with almost no LAM on either electrode. This is in agreement with the earlier observed constant capacity fade rate and matches the literature for comparable stresses [19,38,41]. Regarding the applied low cyclic stresses, primarily parasitic side reactions are assumed to cause SEI and cover layer growth, thereby consuming active lithium [21,39]. Moreover, we suppose the cyclic stresses due to the frequently repeated CUs to be more harmful than the actual cyclic stresses of the investigated test scenario, as reported in literature [56,90].

In contrast, the highway scenario and the 1C/1C cycling trigger a more complex degradation behavior with the capacity of the negative electrode increasing during the first 800 and 600 EFC, respectively, before severe LAM_{NE} sets in. Literature suggests three different explanatory approaches based on observations comparable to ours. Oehler et al. [96] trace this behavior back to the impact of the negative electrode's overhang. They explain that equalization processes lead to charge transfer into the overhang during the pOCV measurement, thus shifting the graphite stage II feature towards higher SOCs, as more charge is required to fully lithiate the negative electrode. This explanation is in alignment with the observations of Li et al. [97,98] for calendar and cycle aging tests, where the same effect occurs at low storage SOC and a small DOD in the low SOC range, whereas it is absent for test cases at higher SOCs. Hence, in the affected test conditions, the overhang has a lower SOC than the active part of the negative electrode and, thus, exhibits a higher tendency for such equalization effects. This matches the lower average SOC during cycling in the highway scenario and 1C/1C test case compared to the commuter scenario, which does not show this effect.

An alternative explanation for the observed gain in capacity of the negative electrode is the local overintercalation of lithium beyond LiC_6 in the active material due to high charge currents during the cycling sequence [99–101], causing increasing capacity of the negative electrode due to deintercalation during the CU. This hypothesis matches the data presented by Li et al. [102], where the same effect occurs at 1C and 2C charge rates while it is absent for lower charge rates.

Apart from that, Guo et al. [103] provide a third theory about the increasing capacity of the negative electrode. They observe an enlarged d-space between the graphite layers, which improves lithium-ion diffusion and increases the amount of intercalable graphite. Moreover, they find a correlation between the applied DOD and the extent of the capacity gain, which is in good agreement with our results. However, this effect cannot be resolved solely based on the electrical measurements and requires post-mortem analysis.

LAM_{PE} is low but present for the highway scenario and the 1C/1C cycling and noticeably coincides with the progression of the resistance increase. Gauthier et al. [21] demonstrate a correlation between LAM_{PE} and resistance growth due to transition metal dissolution and subsequent deposition at the negative electrode, accelerating SEI growth [86] and deterioration of the lithium intercalation kinetics [26]. This is particularly known for nickel-based electrode materials [91,104] and especially promoted by elevated voltage levels [26]. At these elevated voltage levels, the positive electrode is highly delithiated. This distorts the lattice and destabilizes the layered crystal structure, reducing the bonding strength between the transition metal ions and the oxygen, ultimately reducing the mechanical properties of the active material [105–107]. Furthermore, elevated cell potentials promote rock salt formation, decreasing the positive electrode's capacity [108]. In addition, Schuster et al. [26] observe more severe degradation when performing CCCV compared to cycling without CV phase. In contrast, Maheshwari et al. [39] find the opposite behavior. Thus, we assign this effect to the frequently repeated full charges up to 4.19 V, which is in good correlation with literature [109–111]. Compared to the highway scenario, the more pronounced degradation of the positive electrode due to the 1C/1C cycling coincides with the more frequent charge operations to the upper cut-off voltage. Whereas the effect of LAM_{PE} itself is minor, the resulting mechanisms could be crucial influencing factors for the observed subsequent knee point, as suggested in literature [105,112,113].

While initially LLI dominates the degradation behavior presumably due to SEI growth, accelerating LAM_{NE} occurs for the highway scenario around 1400 EFC and for the 1C/1C cycling around 1500 EFC. This is in good correlation with the observed knee point in the capacity losses, indicating a change in the degradation behavior. We assume this to result from aggravated lithium plating, as it coincides with elevated rates of LLI due to the increased consumption of lithium shortly before [89]. The preceding severe degradation of the negative electrode with comparably minor losses of active lithium in the highway scenario poses an increased risk of lithium plating [20,58]. This is not surprising, as a relatively slowly decreasing amount of charge carriers need to be hosted by a significantly reduced amount of active material. In addition, the loss of active material, as well as the progressive formation of cover layers and local passivation of the electrode surface, increase the local current densities and deteriorate the reaction kinetics.

The occurrence of the knee point in the highway scenario seems surprising when comparing the degradation path with the 1C/1C test case. The former employs a power profile representing the necessary drive power, which remains the same throughout the entire experiment, independent of the cell condition. Therefore, with progressive capacity fade due to aging, the utilized voltage window inevitably widens to 2.5 V, which increases the occurring peak currents. In contrast, the applied current in the 1C/1C cycling is constant throughout the entire experiment. Moreover, the voltage window remains fixed to the initial condition, thus causing the applied charge throughput per cycle to decrease over aging. For this reason, voltage ranges below 3.25 V and, thus, low lithiation stages of the negative electrode are avoided. The consideration of the degradation behavior before the knee point, as well as the alterations in the stress factors, suggests the SOC range to be the main degradation-causing stress factor, as the highway scenario initially exhibits a lower degradation rate than the 1C/1C cycling despite higher charge rates and discharge peak currents. Consequently, when avoiding low SOC ranges below 3.25 V such as in the 1C/1C cycling, the cycle life in the highway scenario might even be significantly longer than observed in our experiments.

Table 2
Degradation modes extracted from the vehicle and cell DV curves at a test progress of around 60 000 km.

	LLI	LAM_{NE}	LAM_{PE}	C/C_0
Vehicle	1.9%	−0.5%	0.0%	98.0%
Commuter cell	2.7%	0.0%	0.3%	97.3%

Moreover, the highway scenario and 1C/1C cycling cause severe flattening of the graphite-related features until they completely vanish. This indicates inhomogeneous effects at the negative electrode, either irreversible inhomogeneous degradation [114] or reversible inhomogeneous lithiation [94,115]. The latter is known to be increased by higher cyclic stresses in terms of C-rates and DOD, such as in the two affected test cases [36,61,94]. In conjunction with the observed capacity recoveries during test interruptions, this observation suggests that the identified LLI in the highway scenario and in the 1C/1C cycling could be partially reversible. Taking into account that the inhomogeneous lithiation of the negative electrode is assumed to promote localized lithium plating once a critical level is exceeded [61], the onset of the knee point could be delayed if such inhomogeneities are avoided.

In conclusion, it has to be mentioned that with progressive aging, the degradation mode identification is impeded due to electrode-specific features becoming less pronounced and vanishing towards the end of the experiments. This complicates the precise half-cell alignment and, thus, increases the uncertainty of the presented values towards the end of the test.

3.2. Vehicle perspective

Aging on the vehicle level is assessed to understand degradation under real-world operating conditions. Fig. 5 presents the DV curves on vehicle level in the initial and aged condition at a mileage of around 60 000 km and compares it to the respective DV curves of the commuter scenario. The curves are again aligned by the graphite stage II, indicated by the dashed line. Table 2 summarizes the associated degradation modes.

As depicted in Fig. 5, the dominant degradation mode is LLI, presumably caused by parasitic side reactions such as SEI and cover layer growth. This is not surprising as the test vehicle is parked around 96% of the time, and consequently, calendar aging accounts for the largest share of the overall stress.

No signs of LAM_{NE} are found, even though repeated fast-charging events have been performed. Instead, the negative electrode appears to even slightly increase in capacity, like in the highway scenario and the 1C/1C cycling in the laboratory tests. Yet, it remains to be seen whether this is an actual physical effect as discussed in Section 3.1 or caused by measurement influences, such as temperature effects and the discharge process before the charging measurement, or data post-processing artifacts, such as the smoothing procedure [78,116]. Besides, no inhomogenization effects at the negative electrode are found, as seen by the unaltered graphite phase transition features. The increase of the stage II peak height results from the enhanced superimposition by the M phase as a consequence of LLI. This, again, is in line with the comparably low demanding load spectrum during operation and the long idle sequences in between and matches the

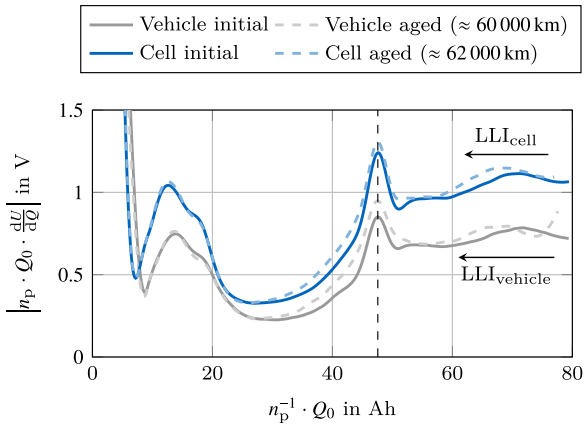


Fig. 5. DVA on vehicle level (gray) in the initial state (solid) and aged condition (dashed) at approximately 60 000 km during 3.5 years of real-life usage in comparison to the cell subjected to the commuter scenario (blue). For better comparability between the vehicle and the cell level, the DV is normalized by the capacity of the parallel connected cells ($n_p = 1$ on cell level and $n_p = 2$ on vehicle level). Equally, the charge Q is normalized by the number of parallel connected cells. Moreover, the DV curves are again aligned to the graphite stage II feature for better recognizability of the degradation modes.

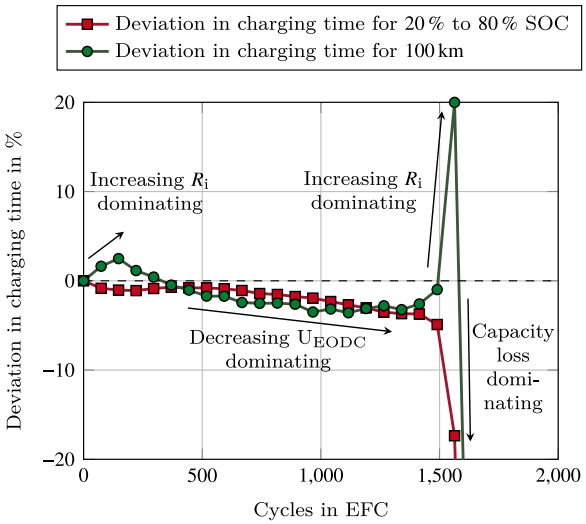


Fig. 6. Deviation in charging time during the fast-charging event of the highway scenario due to battery degradation throughout the aging study. To evaluate the charging time for 100 km, only the initial sequence of the DC fast-charging event is considered. Both cases are normalized to the first fast-charging process.

findings on the cell level well.

The vehicle is often operated between 70% and 100% SOC, thus cycling around the H2 phase of the positive electrode, which harms the active material [117]. Moreover, frequent full charges and parking sequences at high cell voltages promote transition metal dissolution at the positive electrode, as discussed earlier. Despite these operating conditions, Fig. 5 and Table 2 suggest that there is almost no degradation of the positive electrode.

Comparing the stresses with the laboratory tests, the real-world operation of the test vehicle ranges between the commuter and the highway driving pattern, as it incorporates short and long trips with urban, interurban, and highway driving as well as wallbox and fast-charging events. Analyzing the aging behavior reveals that the real-world degradation matches well with the laboratory commuter scenario, while the highway scenario, as well as the 1C/1C cycling, exaggerate the electrode degradation and the inhomogenization of the lithium distribution in the negative electrode. In conclusion, real-world aging seems to be dominated by calendar aging effects despite the versatile ranges of usage of the test vehicle.

3.3. Operational perspective

This section investigates the impacts of the examined aging behavior from a user's perspective. To answer the question of how the observed aging effects, namely capacity loss and resistance increase, affect the user in daily operation, we analyze the influences of cell aging on user-relevant attributes. Therefore, we assess range, charging time, and power limitations during driving, such as the potential of accelerating and regenerative braking. The following considerations are based on the highway scenario because no considerable aging effects occur in the commuter scenario. It is essential to consider that transferring laboratory results to practical vehicle applications is particularly challenging and requires accounting for a multitude of factors. These arise from the dynamics of operating conditions, environmental influences, and cell interconnection and system integration in the vehicle.

3.3.1. Range deterioration

The remaining capacity, as shown in Fig. 3(a), is approximately proportional to the achievable range and, consequently, directly impacts the user, assuming that the BMS does not adjust the available voltage window. The range of the cell subjected to the highway scenario at the end of the OEM's warranty (160 000 km) is reduced by 20 km (7.5%) compared to the initial condition, whereas at the OEM's end of life (EOL) (70% state of health (SOH)) 71 km (26.5%) less can be traveled with a complete discharge. At the end of our test, when the highway cell reached a SOH of around 50%, the achievable range decreased by 132 km. Thus, a noticeable effect for the user only occurs after surpassing the knee point.

The average daily distance traveled by car in the United States is between 47 km [118] and 59 km [119] whereas the European average is 27 km [120]. In Germany on average 22 km are traveled in metropolitan and 37 km in rural areas [121]. Moreover, Bollerslev et al. [122] present data showing that the share of long-distance trips of 250 km and above are below 0.1% in Denmark. Pearre et al. [123] further report that distances of 150 miles (approx. 241 km) on average only occur nine times per year. As the above values represent the high-demanding highway edge case scenario, which exhibits a rather severe degradation behavior, the impact of cyclic aging is minor for most users. Therefore, it can be assumed that the aging-induced range loss does not considerably limit average everyday mobility for most users before reaching the knee point. It should be noted, however, that the achievable range depends on the extractable energy, which is heavily influenced by the performance of the weakest cell. Consequently, a single severely degraded cell can limit the performance of the entire battery pack and, in turn, the achievable range.

3.3.2. Fast-charging capability

In the case of long-distance trips, besides the range, charging time is also crucial. Fig. 6 shows the impact of battery aging on charging time. The red curve represents the deviation of the fast-charging time in the DC charging process between 20% and 80% SOC, while the green curve visualizes the deviation in time to recharge 100 km, both normalized to the first fast-charging event. The data was extracted from every 50th fast-charging cycle and compared to the first one.

The time to perform the entire fast-charging process decreases comparably to the cell capacity. Before the onset of the knee point, it ranges between 34 min and 28 min. This observation can be traced back to three interacting effects. Firstly, the chosen voltage thresholds of the approximated fast-charging profile are reached earlier during the charging process due to the aging-induced internal resistance increase. This causes an earlier current derating than in the initial state, resulting in increased charging time. Secondly, the amount of energy to be

recharged also decreases as capacity fades over aging, which reduces the charging time. Thirdly, the end of discharge (EODC) voltage, and therefore the starting point of the fast-charging event, shifts towards lower voltage levels as a result of the decreasing capacity, providing a larger voltage window before reaching the upper cut-off voltage.

However, a different behavior becomes apparent when considering only the first part of the fast-charging event, e.g., in order to recharge 100 km. Initially, the charging time increases since the resistance increase is more dominant than the capacity fade in the first cycles. After the second CU, the resistance becomes almost constant while the capacity further decreases and starts dominating the behavior. Therefore, the EODC voltage and, thus, the starting point of the fast-charging declines. Hence, a tipping point is reached, leading to a decrease in charging time. Towards the end of the test, the charging time for 100 km increases again as a consequence of the substantial resistance growth. As a further result of the resistance increase, the recharged energy declines substantially in the last data point and, with it, the charging time.

In contrast to the entire fast-charging (red), the 100 km charging (green) represents a constant amount of energy throughout the entire test. Nevertheless, after the initial increase in charging time, it exhibits a more significant decrease. Hence, the charging time decreases in the lower and medium SOC region, as the charging power derating takes place later. It should be noted that in the presented data, no sophisticated aging-related charging strategy is considered, as it might be implemented in the vehicle. Moreover, the charging performance of the battery pack is impacted by the cell interconnection and the specific states of individual cells. Uneven splitting of the charging current among parallel-connected cells, caused by differences in SOC, resistances, or temperatures, can result in variations in local charging performance and associated stresses across the battery pack. Similar to the range, the overall charging performance is further limited by the weakest cell, as power is restricted to prevent overloading specific cells.

3.3.3. Power limitations

Moreover, we determine the cell's power limits for the measured aging behavior to quantify the impact of the aging-induced internal resistance increase on the performance capability. The power applied to or drawn from a cell during operation is limited by the BMS to avoid safety or aging critical conditions. This is achieved by setting upper and lower boundaries to the battery voltage. The cell power limits are reduced when the cell's internal resistance increases to avoid violating voltage and thermal limits.

The power limits that prevent violating the cell's voltage limits are calculated using Eqs. (1) and (2), with $P_{\mathrm{ch,lim}}$ and $P_{\mathrm{dch,lim}}$ denoting the charging and discharging limits, U_{EOC} and U_{EODC} the end of charge and discharge voltage, U_{OCV} the open circuit voltage, and $R_{\mathrm{DC,ch}}$ and $R_{\mathrm{DC,dch}}$ the cell's internal resistance in charging and discharging direction.

$$P_{\mathrm{ch,lim}} = \frac{U_{\mathrm{EOC}} \cdot (U_{\mathrm{EOC}} - U_{\mathrm{OCV}})}{R_{\mathrm{DC,ch}}} \tag{1}$$

$$P_{\mathrm{dch,lim}} = \frac{U_{\mathrm{EODC}} \cdot (U_{\mathrm{EODC}} - U_{\mathrm{OCV}})}{R_{\mathrm{DC,dch}}} \tag{2}$$

The upper and lower voltage limits are set to 3.25 V and 4.19 V. The open circuit voltage was approximated based on the pOCV measurement, and the resistances correspond to the 20 s 1 C pulses in charge and discharge direction, as presented in Section 3.1.1.

In Fig. 7, the calculated power limits are shown against EFC for the highway scenario. Note that these results represent theoretical limits only taking into account the electrical behavior and neglecting further limiting factors, such as the cell temperature, the maximum current, and alterations in the SOC. In addition, the gray shaded area depicts the peak power limits of the *VW ID.3* motor of 150 kW, which equals 0.69 kW on cell level. The positive motor power represents the generative operation during regenerative braking. Additionally, the

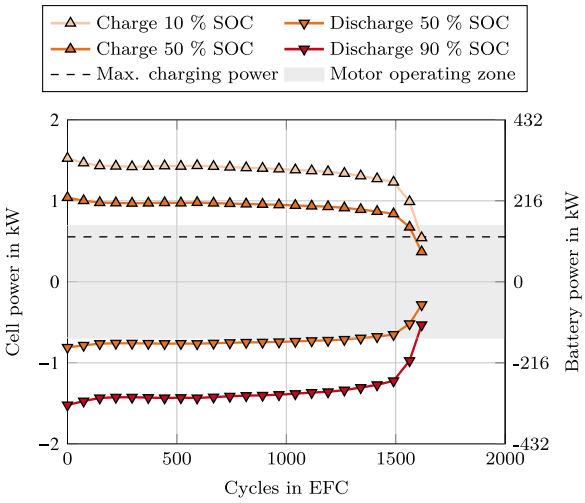


Fig. 7. Impact of resistance increase on the theoretical battery cell and pack power limits. The power values have been calculated using the 20 s 1 C pulses in charge and discharge direction at various SOC levels. The gray shaded area represents the motor operating zone neglecting battery limitations. The maximum charging power of the 100 kW DC fast-charging is indicated by the dashed line.

maximum charging power at the start of fast-charging of 100 kW is indicated by the dashed line. Throughout the aging study, the power limits tighten due to the internal resistance increase as the battery ages. However, the drive train power limits are only constrained by the battery cell after the knee point at around 1500 EFC. Apart from that advanced aging state, the electric motor and not the battery restricts the retrievable drive train power in charge and discharge direction.

The results show that the battery operation, like driving, regenerative braking, and charging, will only be slightly affected by the aging-induced resistance increase from an electrical point of view. However, a higher resistance also impacts thermal losses, which may require tightening the power limits after prolonged operation for thermal reasons. Yet, the measured cell surface temperatures in our laboratory tests, as well as in the vehicle operation, do not suggest any severe thermal alterations [40]. The achievable charging speed in the vehicle may, however, be reduced actively through the BMS due to a higher risk of lithium plating throughout the aging progress, which cannot be deduced from these measurements. Nonetheless, the results suggest that the impact on the perceived performance is minor.

3.3.4. Lifetime prognosis

For automotive applications, the EOL of a LIB is typically defined at 70 to 80% of its initial capacity [24,34,50] or at 150 to 200% of its initial resistance, respectively [50,71]. In literature, mileages between 150 000 and 250 000 km are commonly expected until this point [124–127]. Volkswagen defines the EOL of their BEVs at 70% SOH and guarantees a mileage of 160 000 km [128].

So far, the aging effects have been analyzed in terms of EFC. However, the more relevant consideration for the customer is aging over mileage. The conversion from cycles to mileage is straightforward for the commuter and the highway scenario since these load patterns originate from driving profiles recorded with the test vehicle in public road traffic, thus providing the related energy consumption. For the less challenging commuter driving pattern, the specific energy consumption on the cell level amounts to 0.79 Wh/km. The more demanding highway driving pattern exhibits a specific energy consumption of 0.92 Wh/km on the cell level. The resulting SOH curves over mileage are shown in Fig. 8. The mileage covered by the OEM warranty corresponds to 448 cycles in the highway scenario and 5992 cycles in the commuter scenario. At this point, the SOH of the highway cell has reached 92.2%, while the commuter cell is still above 95%. During the test period, only

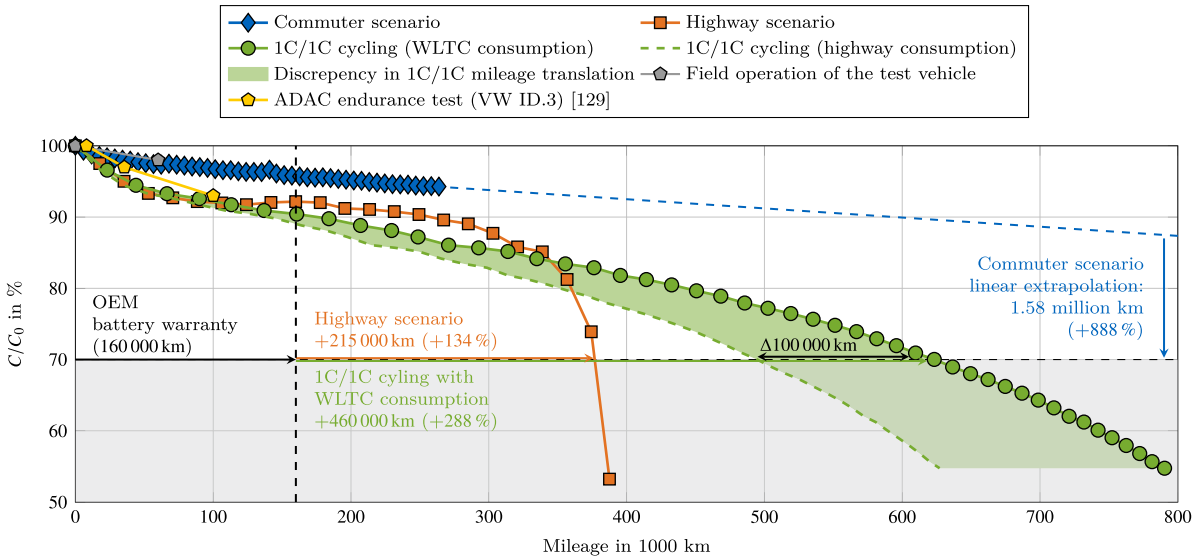


Fig. 8. Cycle life performance under the three examined test scenarios transferred to mileage in comparison to the OEM's battery warranty. The green curve of the 1C/1C cycling represents the obtained mileage based on the WLTC energy consumption of the *VW ID.3*. The green shaded area and the deviation of approximately 100 000 km at 70% SOC result from the imprecision of the mileage transfer approach. The lower boundary corresponds to the mileage based on the energy consumption of the highway scenario. The yellow curve represents the aging data from the VW ID.3 measured by the General German Automobile Association [129], whereas the gray curve corresponds to the test vehicle.

the highway cell has reached its EOL at nearly 1600 EFC, which equates to approximately 375 000 km. Thus, the OEM's cyclic warranty is more than doubled, even under the challenging highway driving profile with frequent fast-charging events and 1.5 years of continuous cycling.

The translation of EFC into mileage is more complex for the 1C/1C test case as this synthetic load profile does not correspond to an actual driving operation. Therefore, it cannot be transferred directly into a related mileage. For this reason, we suggest a conversion approach based on the energy consumption under standard conditions like the WLTC, which is provided by the OEM. Based on the given WLTC range of 402 km [130], an averaged specific energy consumption of 0.75 Wh/km on cell level is obtained. Multiplying this energy consumption by the measured energy throughput during the aging test provides the green SOH trajectory in Fig. 8, yielding a corresponding mileage of approximately 620 000 km until EOL.

Besides altering the degradation behavior, this approach demonstrates a further problem of synthetic test scenarios when aiming to assess the battery lifetime in terms of vehicle-relevant characteristics such as mileage. By translating the energy throughput into the respective mileage, the underlying loads are neglected. Consequently, when using this consumption-based approach, only different energy consumptions are considered, but not the corresponding stress factors under various usage patterns. To better evaluate the sensitivity of the mileage projection concerning the chosen driving condition, the SOH trajectories for the 1C/1C aging curve with the commuter and highway energy consumptions are determined as well. The result is visualized as the shaded area in Fig. 8, with the upper boundary related to the WLTC consumption, the lower boundary (dashed line) connected to the highway consumption, and the trajectory with the commuter consumption lying in between. At EOL (70% of the initial capacity), an uncertainty of around 100 000 km results from the mileage transfer approach.

The mileage reached in the highway scenario can be translated into a service life of almost 20 years, assuming an annual driving range of around 20 000 km. However, when extrapolating the service life, it should be noted that several effects impacting the long-term behavior come into play. On the one hand, the impact of calendar aging is strongly underestimated in the laboratory tests compressed in time. Nevertheless, this effect especially needs to be considered when extrapolating the service life since it appears to dominate the degradation behavior in real-world operation, as shown in Section 3.2. On the other hand, lithium inhomogenization in the negative electrode during continuous cycling is assumed to accelerate cell degradation and to be one of the origins of the knee point in the highway scenario, compare Section 3.1. In real-world operation, however, no indications of inhomogenization have been found, and the battery pack exhibits a linear-like capacity decrease instead (Section 3.2). This leads to the conclusion that in the absence of such significant inhomogenization effects, a knee point might occur considerably delayed.

Taking into account that the vehicle exhibits similar degradation behavior to the cell subjected to the commuter scenario, mostly dominated by calendar aging, we assume a longer lifetime in real-world operation than in the highway scenario. Moreover, no considerable impacts on the user have been found before the knee point, even in the highly demanding highway scenario, see Section 3.3. To further evaluate the cell degradation behavior measured in the laboratory, the data is compared to a second real-world vehicle test conducted by the ADAC [129]. In this endurance test, the test vehicle was subjected to a usage pattern similar to the presented highway profile, including large DODs, highly dynamic driving operations, frequent charging events to 100% SOC, and a fast-charging share of almost 40%. As shown in Fig. 8 the aging behavior matches the highway scenario well.

The comparison of the laboratory data and the vehicle data supports the assumption that real-world battery degradation might lie somewhere in between the two presented edge cases for most use cases. Hence, to answer the question of which mileage is to be expected in the field application in the best case, we linearly extrapolate the measured cell degradation of the commuter cell. Linearizing the respective measurement data yields a degradation rate $\Delta SOH_{\mathrm{EFC}}$ of $-0.49\%/100\,\mathrm{EFC}$ and ΔSOH_{km} of $-0.019\%/1000\,\mathrm{km}$ leading to a cycle life prediction of more than 1.5 million km before reaching the OEM's EOL criterion. It has to be taken into account that in the long term, even the minor degradation effects observed in the commuter scenario and the real-world operation can trigger severe aging mechanisms at some point, which is why this best-case prediction may not be reached in reality.

4. Summary and conclusions

In this work, we present a comprehensive cycle aging characterization of automotive-grade battery cells from the *Volkswagen ID.3 Pro Performance* analyzing the occurring aging effects and underlying degradation modes on the cell and vehicle level, as well as assessing the aging impacts from a user perspective. We answer how state-of-the-art lithium-ion battery cells degrade under two realistic edge-case operating scenarios in contrast to an accelerated synthetic test case and which modes and mechanisms dominate the degradation process. Based on these findings, we assess the comparability of aging tests under accelerated synthetic load conditions with realistic ones, as well as the transferability of single-cell laboratory tests to the vehicle level throughout more than 60 000 km of operation in the field. The following three central findings can be concluded:

(1) Remarkable cycle life performance.
The tested battery cells substantially outperform the manufacturer's warranty. Even under demanding operating conditions such as highly dynamic long-distance highway driving and repeated fast-charging events, the cells complete nearly 400 000 km, 2.5 times the guaranteed mileage. Moreover, the cells pass more than 1100 fast-charge cycles from 20 % to 80 % SOC before reaching EOL. No severe cyclic aging mechanisms were found for the commuter driving pattern, suggesting an aging behavior dominated by calendar aging and a theoretical linearly extrapolated mileage of over 1.5 million km.

(2) Manipulation of the degradation behavior through accelerated testing.
Accelerated synthetic load conditions save considerable amounts of testing time. While 1C/1C CC cycles allow for the complete experimental coverage of the given OEM's cyclic aging warranty of 160 000 km within 14 weeks, realistic load scenarios take up to 1.5 years. However, this acceleration is accompanied by the manipulation of the occurring degradation behavior. The commuter cell appears to primarily exhibit LLI, whereas almost no electrode degradation has been found. In contrast, the highway scenario additionally causes significant LAM_{NE} whereas in the 1C/1C test case LAM_{NE} even dominates the degradation behavior and noticeably more LAM_{PE} occurs, indicating that also the underlying aging mechanisms differ. More importantly, while the 1C/1C test case initially induces faster degradation, it outperforms the highway scenario in the long term by almost 30%, highlighting the limited transferability between different stress factors, with the DOD being one main influencing factor.
Another crucial mismatch between the test scenarios is the considerably inhomogeneous condition of the negative electrode. This can be attributed to inhomogeneous degradation as well as potentially inhomogeneous lithium distribution within the electrode. The latter results from the omission of rest periods between the cycles. Indeed, this fact enables a significant saving of test time and is commonly employed for this reason. However, it suppresses essential cell relaxation and compensation mechanisms. This effect is exacerbated by constant current cycling compared to dynamic real-world load spectra, but it is not apparent in the less demanding commuter scenario. While this inhomogeneous lithiation of the negative electrode is a reversible phenomenon at first, it is assumed to irreversibly affect cell degradation in the long term as it promotes lithium plating. For this reason, we suppose that the knee point in the highway scenario occurs prematurely compared to real-world operation.

(3) Uncertainty in the transferability from laboratory to vehicle level.
The reliability of aging data gained from synthetic accelerated tests used for aging characterization and lifetime prognosis in the automotive application is questionable. Battery pack aging in the field application primarily exhibits LLI, while almost no electrode degradation is found. As a result, the degradation behavior corresponds well with that of the commuter scenario in the laboratory tests. In contrast, the other two test cases reveal a significant discrepancy compared to the real-world aging behavior. Since the vehicle rests approximately 96% of the time in the real-world application, no noticeable inhomogenization of the negative electrode's lithiation has been found. Therefore, one limiting key factor for transferability is the omission of rest periods. Other than that, the transfer from the laboratory level into the practical application is impacted and limited by various other factors. While the operating conditions under laboratory environments are commonly abstracted and static, the real-world conditions are highly dynamic and diverse: complex load profiles, versatile partial cycles, and long-term usage with superimposed cycle and calendar aging occur, which are often not fully replicated in laboratory tests. Furthermore, environmental factors such as temperature, humidity, and mechanical stress significantly impact battery performance. On a system level, further aspects such as cell interconnection and unequal current distribution in parallel connected cells, as well as interventions by the BMS and thermal management, come into play, which is typically absent in lab-based investigations. Moreover, scaling effects complicate the transferability from individual cells to modules and packs, as phenomena such as heat generation and internal resistance do not scale linearly. To ensure a reliable transfer of laboratory results to vehicle-level applications, realistic tests, comprehensive models, and integrated approaches that account for external factors and dynamic processes are essential.

Translating these findings into a practical application implies several suggestions. Firstly, a less conservative implementation of the battery system and its operating strategy is conceivable based on the observed excellent cycle life performance. This includes, e.g., more time efficient charging profiles, which would be directly noticeable for the user in a shorter charging time. *Volkswagen* has already taken this measure in mid-2021 through an over-the-air software update, raising the maximum charging power from 100 kW to 120 kW for the investigated test vehicle [131]. Moreover, as our results identify the DOD and cycling in the low SOC range as dominating stress factors, avoiding cell potentials below 3.2 V and incorporating sufficient relaxation phases appear promising to further extend the battery's lifetime. Secondly, also on the test level, particular attention should be paid to specifying the predominant stress factors in accordance with the related application to avoid severe manipulation of the degradation behavior. In addition, dedicated rest periods should be incorporated into the test design to take relaxation effects into account in order to improve the transferability of the experimental data. Despite the unexpected high cycle life, it has to be noted that during the test period of more than three years, cell technology has further advanced. For this reason, even better lifetime performance is expected in the current generations of LIBs.

We suggest further in-depth investigations on automotive-grade vehicle cells. Our future work will investigate relaxation effects during rest periods to improve the transferability between accelerated and unaccelerated aging tests. In addition, the calendar aging behavior needs to be assessed to create a complete understanding of the cell's aging behavior. Finally, post-mortem analysis could help validate the identified aging mechanisms.

CRediT authorship contribution statement

Markus Schreiber: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation,

Conceptualization. **Kareem Abo Gamra:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation. **Philip Bilfinger:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation. **Olaf Teichert:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation. **Jakob Schneider:** Writing – review & editing, Writing – original draft, Visualization, Investigation. **Thomas Kröger:** Writing – review & editing, Writing – original draft, Visualization, Investigation. **Nikolaos Wassiliadis:** Writing – review & editing, Writing – original draft, Methodology. **Manuel Ank:** Writing – review & editing, Writing – original draft. **Marcel Rogge:** Writing – review & editing, Writing – original draft, Visualization, Investigation. **Jan Schöberl:** Writing – review & editing, Writing – original draft. **Philipp Rosner:** Writing – review & editing, Writing – original draft. **Cristina Grosu:** Writing – review & editing, Writing – original draft. **Andreas Jossen:** Writing – review & editing, Supervision, Funding acquisition. **Markus Lienkamp:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Cycle aging test protocols

Tables A.1–A.3 summarize the cycle aging test procedures for the three test scenarios. The dynamic load profiles were imported as $2 \times n$ tables containing time and power vectors, stored in a .txt file.

Appendix B. Load profiles

The load profiles of the three test scenarios depicted in Fig. B.1 in terms of voltage and power are. Moreover, Fig. B.2 visualizes the fast-charging profile applied in the highway scenario.

Appendix C. Incremental capacity analysis

ICA is utilized to validate the degradation modes interpreted from the DVA. For this purpose, Fig. C.3 presents the ICA spectra of the three test cases, highlighting the observed degradation modes based on the curve alterations.

Table A.1
Cycle aging test protocol for the commuter scenario.

#	Command	Parameter	Termination	Sampling
1	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
2	Charge	CP @ $P_{11\,\mathrm{kW}}$[a]	$U > U_{\mathrm{SOC80}}$[b]	$t = 10\,\mathrm{s}$
3	Charge	CV @ U_{SOC80}	$I < C/20$	$t = 10\,\mathrm{s}$
4	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
5	Loop start	–	–	–
6	Discharge	Commuter pattern[c]	$Q > Q_{\mathrm{cycle}}$	$t = 2\,\mathrm{s}$
7	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
8	Charge	CP @ $P_{11\,\mathrm{kW}}$	$U > U_{\mathrm{SOC80}}$	$t = 10\,\mathrm{s}$
9	Charge	CV @ U_{SOC80}	$I < C/20$	$t = 10\,\mathrm{s}$
10	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
11	Loop end	Count = 200	–	–
12	Pause	–	$t > 900\,\mathrm{s}$	$t = 10\,\mathrm{s}$

[a] The charging power $P_{11\,\mathrm{kW}}$ equals approximately 50.9 W.
[b] $U_{\mathrm{SOC80}} = 4.019\,\mathrm{V}$; cell voltage measured in the vehicle at 80% SOC.
[c] The commuter driving pattern is embedded as a table.

Table A.2
Cycle aging test protocol for the highway scenario.

#	Command	Parameter	Termination	Sampling
1	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
2	Charge	CP @ $P_{11\,\mathrm{kW}}$[a]	$U > U_{\mathrm{SOC100}}$[b]	$t = 10\,\mathrm{s}$
3	Charge	CV @ U_{SOC100}	$I < C/20$	$t = 10\,\mathrm{s}$
4	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
5	Loop start	–	–	–
6	Discharge	Highway pattern[c]	$Q > Q_{\mathrm{cycle,1}}$[d]	$t = 2\,\mathrm{s}$
7	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
8	Charge	DC fast-charging[e]	$U > U_{\mathrm{SOC80}}$[f]	$t = 10\,\mathrm{s}$
9	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
10	Discharge	Highway pattern	$Q > Q_{\mathrm{cycle,2}}$[g]	$t = 2\,\mathrm{s}$
11	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
12	Charge	CP @ $P_{11\,\mathrm{kW}}$	$U > U_{\mathrm{SOC100}}$	$t = 10\,\mathrm{s}$
13	Charge	CV @ U_{SOC100}	$I < C/20$	$t = 10\,\mathrm{s}$
14	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
15	Loop end	Count = 50	–	–
16	Pause	–	$t > 900\,\mathrm{s}$	$t = 10\,\mathrm{s}$

[a] The charging power $P_{11\,\mathrm{kW}}$ equals approximately 50.9 W on cell level.
[b] $U_{\mathrm{SOC100}} = 4.19\,\mathrm{V}$; cell voltage measured in the vehicle at 100% SOC.
[c] The highway driving pattern is embedded as a table.
[d] $Q_{\mathrm{cycle,1}}$ is the amount of charge of the first partial cycle.
[e] The DC fast-charging profile is embedded as a table.
[f] $U_{\mathrm{SOC80}} = 4.019\,\mathrm{V}$; cell voltage measured in the vehicle at 80% SOC.
[g] $Q_{\mathrm{cycle,2}}$ is the amount of charge of the second partial cycle.

Table A.3
Cycle aging test protocol for the 1C/1C test case.

#	Command	Parameter	Termination	Sampling
1	Pause	–	$t > 60\,\mathrm{s}$	$t = 10\,\mathrm{s}$
2	Loop start	–	–	–
3	Charge	CC @ 1C	$U > U_{\mathrm{SOC100}}$[a]	$t = 10\,\mathrm{s}$
4	Charge	CV @ U_{SOC100}	$I < C/20$	$t = 10\,\mathrm{s}$
5	Pause	–	$t > 600\,\mathrm{s}$	$t = 10\,\mathrm{s}$
6	Discharge	CC @ 1C	$U < U_{\mathrm{SOC0}}$[a]	$t = 2\,\mathrm{s}$
7	Pause	–	$t > 600\,\mathrm{s}$	$t = 10\,\mathrm{s}$
8	Loop end	Count = 100	–	–
9	Pause	–	$t > 900\,\mathrm{s}$	$t = 10\,\mathrm{s}$

[a] $U_{\mathrm{SOC100}} = 4.19\,\mathrm{V}$; cell voltage measured in the vehicle at 100% SOC.
[b] $U_{\mathrm{SOC0}} = 3.25\,\mathrm{V}$; cell voltage measured in the vehicle at 0% SOC.

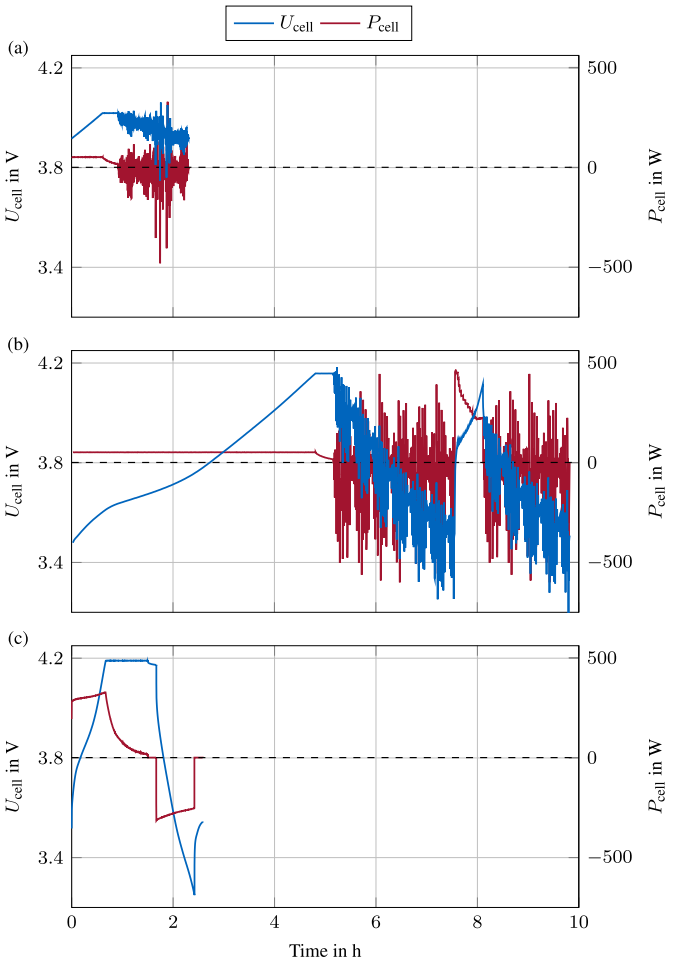


Fig. B.1. Load profiles of the three investigated test cases visualized by cell voltage and cell power: (a) commuter scenario, (b) highway scenario, (c) 1C/1C cycling.

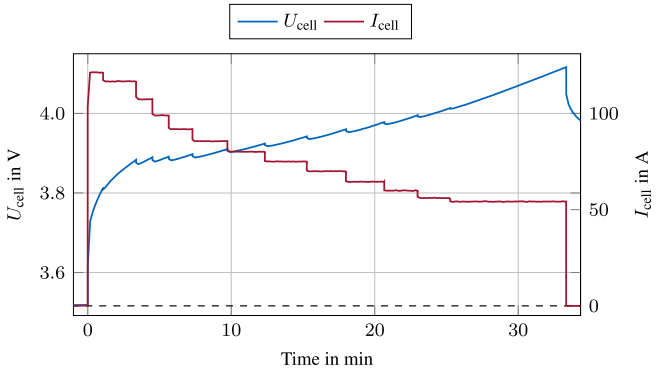


Fig. B.2. Fast-charging profile approximating the measured vehicle fast-charging event applied in the highway scenario, visualized by cell voltage and cell current.

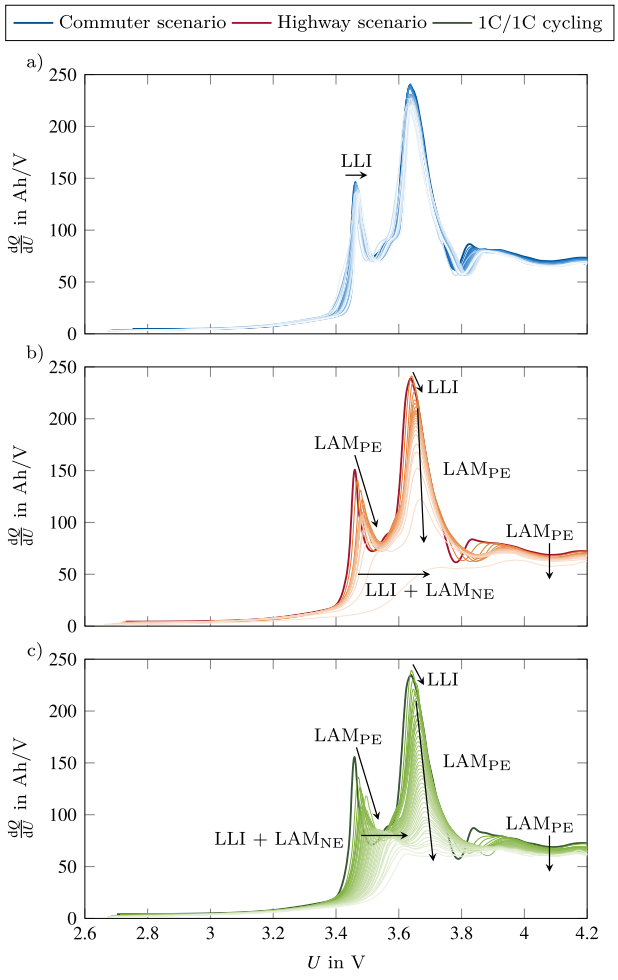


Fig. C.3. Incremental capacity analysis of the examined test cases: (a) commuter scenario, (b) highway scenario, (c) 1C/1C cycling, from dark (pristine condition) to bright (aged condition).

Data availability

We grant everyone unrestricted access to our measurement data. Over 25 GB of measurement data presented in the article are provided as an open source, accessible via mediaTUM: https://doi.org/10.14459/2024mp1748915.

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